Yishan Wang

y.wang18@tue.nl | Eindhoven, Netherlands
yishani.github.io | GitHub | Google Scholar | LinkedIn

RESEARCH INTERESTS

Audio-language models; multimodal learning; voice agents and discourse; agentic learning; self-improving systems; AI for scientific discovery.

EDUCATION

Eindhoven University of Technology June 2025 - present
PhD research in multimodal AI agents *Netherlands*

- Department of Industrial Design. Advisors: Aaqib Saeed and Mathias Funk. Expected completion: May 2029.

University of Amsterdam Sept. 2022 - Sept. 2024
MSc Brain and Cognitive Sciences (Research) *Netherlands*

- **Research placements:** Neural representations and brain-behavior correspondence at CAVA Lab, UvA; implicit linguistic bias in language models at VU Amsterdam.
- **Selected coursework:** Advanced Topics in Philosophy of Language, Computational Cognitive Neuroscience, Research Design & Statistics, Neuroimaging.

Peking University Sept. 2018 - July 2022
BA Philosophy *China*

- **Selected coursework:** Philosophy of Language, Mathematical Logic, Epistemology, Cognitive Psychology, Experimental Psychology, Probability and Statistics.

PUBLICATIONS

[J] Peer-reviewed journal article; [C] Peer-reviewed conference paper; [P] Preprint.

[P] **Yishan Wang**, Ran Piao, Mathias Funk, and Aaqib Saeed. SimAuthor: Harnessing Foundation Models for Persistent Scientific Simulator Authoring. *arXiv preprint*, arXiv:2610.06257, 2026.

[J] **Yishan Wang**, Tsai-Ning Wang, Mathias Funk, and Aaqib Saeed. StethoLM: Audio Language Model for Cardiopulmonary Analysis Across Clinical Tasks. *Transactions on Machine Learning Research (TMLR)*, 2026. Presented at ICML 2026, Journal-to-Conference track. [Code] [Model] [Dataset]

[C] **Yishan Wang**, Amanda Cercas Curry, and Flor Miriam Plaza-del-Arco. Your Mileage May Vary: How Empathy and Demographics Shape Human Preferences in LLM Responses. *Findings of the Association for Computational Linguistics: EMNLP*, pp. 17257-17270, 2025.

[C] **Yishan Wang**, Pia Sommerauer, and Jelke Bloem. The Negation Bias in Large Language Models: Investigating Bias Reflected in Linguistic Markers. *Conference on Language Modeling (COLM)*, 2025.

OTHER RESEARCH EXPERIENCE

Independent Researcher Oct. 2024 - May 2025
In collaboration with MilaNLP, Bocconi University *Remote*

- Studied demographic and empathy-related variation in preferences for LLM responses with Flor Miriam Plaza-del-Arco and Amanda Cercas Curry; published in EMNLP Findings 2025.

Research Intern Mar. 2024 - Sept. 2024
Computational Linguistics & Text Mining Lab, Vrije Universiteit Amsterdam

- Investigated implicit bias in masked and autoregressive language models through linguistic markers and social-psychological frameworks; developed dataset curation and evaluation procedures.

Supervisors: Pia Sommerauer and Antske Fokkens.

Research Intern Feb. 2023 - Oct. 2023
Cognitive AI & Vision Amsterdam (CAVA) Lab, University of Amsterdam

- Mapped deep neural network embeddings to human behavioral and brain data using representational similarity analysis, Python, and PyTorch.

Supervisor: H. Steven Scholte.

ACADEMIC SERVICE

Conference reviewer: NeurIPS 2026; ICLR 2027; AAAI-27.

SKILLS

Programming: Python (PyTorch, NumPy, pandas), R, LaTeX.
Methods: Multimodal modeling and evaluation, dataset curation, experimental design, statistical analysis, representational similarity analysis.
Languages: Chinese (native), English (fluent).